Project Title: Second Serve – AI-Driven Circular Food Redistribution Platform

Problem Statement

Urban cities generate significant quantities of surplus food daily, while vulnerable populations, animal shelters, and small farmers face resource shortages. Restaurants lack structured systems to redistribute excess food safely, and NGOs do not have real-time visibility of available surplus. Additionally, food safety monitoring and post-expiry handling remain inefficient, leading to avoidable waste and environmental harm.

There is a need for an AI-driven, real-time platform that enables intelligent surplus redistribution, freshness prediction, and multi-stage resource utilization within urban ecosystems.

Motivation

- Reduce food waste and landfill methane emissions
- Address urban hunger and food insecurity
- Support animal shelters with safe surplus food
- Promote compost-based support for farmers
- Strengthen circular economy practices in smart cities
- Enable data-driven redistribution rather than manual coordination
- Integrate AI into sustainability and social welfare systems
- Contribute to structured, technology-enabled urban well-being

Application

Target Users

- Restaurants and cafes
- NGOs and food banks
- Low-income individuals
- Animal shelters
- Farmers

- Municipal waste authorities

Deployment Environment

- Urban smart cities (initial pilot: Mumbai)
- Corporate food courts
- Event venues
- College campuses

Real-World Workflow

1. Restaurant uploads surplus food details (type, quantity, preparation time).
2. AI model predicts safe consumption duration.
3. NGOs and individuals receive notifications based on proximity and urgency.
4. If unclaimed within the safe window:
   - Food is reclassified for animal shelters.
5. If unsuitable for animals:
   - Redirected for composting and fertilizer use by farmers.

This establishes a closed-loop redistribution system powered by AI.

Proposed Method (GenAI + ML Model Training)

Second Serve integrates Generative AI and Machine Learning model training to power intelligent decision-making.

1. ML-Based Freshness Prediction Model

We will train supervised machine learning models using historical food spoilage data.

Model inputs:

- Food category
- Preparation time
- Storage conditions
- Weather data
- Past spoilage timelines

Model output:

- Predicted remaining edible time (in hours)
- Risk classification (Safe / Soon Expiring / Expired)

This process of training an ML model involves:

- Feeding historical labelled data
- Optimizing prediction accuracy
- Validating using test datasets
- Continuously improving performance with new data

2. Intelligent Matching Algorithm

Using optimization algorithms and ML ranking systems, the app will:

- Match surplus food with nearest NGO
- Prioritize high-need zones
- Minimize delivery distance and time

3. Generative AI Layer

GenAI components will:

- Generate automated notifications
- Summarize surplus data into impact dashboards
- Create real-time sustainability insights
- Power a chatbot for NGO coordination and onboarding

4. Dynamic Reclassification Engine

An ML classification model will automatically transition food into:

- Human-safe category
- Animal-safe category
- Compost-grade category

This ensures efficient and ethical redistribution.

Datasets / Data Sources

- Historical food shelf-life datasets
- FSSAI food safety guidelines

- Restaurant-provided surplus logs
- Weather API data
- NGO geolocation database
- Animal dietary safety datasets
- Municipal organic waste data

Data types include:

- Structured tabular data
- Image uploads (optional computer vision module)
- Geospatial data
- Time-series data

Experiments

Model Evaluation Surveys and Checks for the Code

- Accuracy testing of freshness prediction model
- Classification performance for spoilage categories
- Code validation and debugging checks
- Cross-validation using test datasets
- NGO feedback surveys on redistribution efficiency
- Pilot testing comparison (manual vs AI-driven matching)
- Waste reduction percentage measurement
- Time-to-redistribution analysis

Novelty and Scope to Scale

Key Innovation ideas

- AI-powered shelf-life prediction
- Automated circular redistribution logic
- Multi-tier waste utilization (Human → Animal → Compost)
- ML-driven adaptive learning system
- Sustainability analytics dashboard

- Integration of social welfare + AI decision intelligence

Scalability Potential

- Expansion to Tier 2 and Tier 3 cities
- Integration with CSR initiatives and corporate cafeterias
- Partnerships with municipal smart-city programs
- Deployment in wedding/event catering ecosystems
- Collaboration with government food security schemes
- Integration with national sustainability dashboards
- Extension to grocery chains and supermarket surplus
- API-based integration with existing food delivery platforms
- Expansion into disaster-relief redistribution networks

Long-Term Vision

Second Serve is not merely a food redistribution platform. It represents a citizen-centric AI system designed to strengthen community well-being.

By intelligently connecting surplus resources with areas of need, the platform supports:

- Social stability
- Resource equity
- Environmental sustainability
- Support systems for vulnerable populations
- Strengthening of the economic and social support pillars of society

The long-term goal is to build an AI-enabled ecosystem that enhances national well-being by reducing waste, supporting citizens, and enabling responsible urban resource management.